

Jinlin Lai

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Positions

Basis AI Institute

SUMMER RESEARCH INTERN

New York, NY

June 2025 - August 2025

Flatiron Institute, Simons Foundation

SUMMER PRE-DOCTORAL RESEARCHER

New York, NY

May 2024 - August 2024

Dolby Laboratories Inc.

ATG IMAGING RESEARCH INTERN

Sunnyvale, CA

June 2023 - August 2023

Education

University of Massachusetts Amherst

PH.D. IN COMPUTER SCIENCE

Amherst, MA, United States

Aug 2020 - June 2026

- Advisor: Dr. Daniel R. Sheldon
- Area: Probabilistic machine learning, computational statistics and Bayesian inference

University of Massachusetts Amherst

M.S. IN COMPUTER SCIENCE

Amherst, MA, United States

Aug 2020 - Feb 2024

Tsinghua University

B.ENG. OF COMPUTER SCIENCE AND TECHNOLOGY

Beijing, China

Aug 2016 - June 2020

- Minor in Finance and Entrepreneurship

Publications

PREPRINT

Jinlin Lai, Charles Margossian, Daniel Sheldon. (2026). Corrected Integrated Laplace Approximation for Bayesian Inference in Latent Gaussian Models. arXiv preprint arXiv:2605.20345. [link]

CONFERENCE

Max Hamilton, **Jinlin Lai**, Daniel Sheldon, Subhansu Maji. (2026). Scalable Model-Assisted Multi-Target Estimation in Large Image Collections. UAI 2026. To appear.

Jinlin Lai, Antonio Ricardo Linero, Yuling Yao. (2026). Predictive variational inference: Learn the predictively optimal posterior distribution. In Proceedings of the 43rd International Conference on Machine Learning (ICML), Seoul, South Korea. PMLR 306, 2026. To appear. [link]

Max Hamilton*, **Jinlin Lai***, Wenlong Zhao, Subhansu Maji†, Daniel Sheldon†. (2025). Active Measurement: Efficient Estimation at Scale. In Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS). [link]

Jinlin Lai, Justin Domke, Daniel Sheldon. (2024). Hamiltonian Monte Carlo Inference of Marginalized Linear Mixed-Effects Models. In Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, Canada. [link]

Jinlin Lai, Anustup Choudhury, Guan-Ming Su. (2024). Outdoor Scene Relighting with Diffusion Models. In Proceedings of the 27th International Conference on Pattern Recognition (ICPR), Kolkata, India. [link]

Jinlin Lai, Javier Burrone, Hui Guan, Daniel Sheldon. (2023). Automatically Marginalized MCMC in Probabilistic Programming. In Proceedings of the 40th International Conference on Machine Learning (ICML), Honolulu, Hawaii, USA. PMLR 202, 2023. [link]

Jinlin Lai, Justin Domke, Daniel Sheldon. (2022). Variational Marginal Particle Filters. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS) 2022*, Valencia, Spain. PMLR: Volume 151. [link]

Haowen Xu, Wenxiao Chen, **Jinlin Lai**, Zhihan Li, Youjian Zhao, Dan Pei. 2020. Shallow VAEs with RealNVP Prior can Perform as Well as Deep Hierarchical VAEs. ICONIP.

WORKSHOP

Aditya Parashar*, Aditya Vikram Singh*, Avinash Amballa*, **Jinlin Lai**, and Benjamin Rozonoyer. (2024). "Quasi-random Multi-Sample Inference for Large Language Models." In *Frontiers in Probabilistic Inference: Learning meets Sampling*. [link]

Jinlin Lai, Daniel Sheldon. 2022. Automatic Inference with Pseudo-Marginal Hamiltonian Monte Carlo. ICML workshop Beyond Bayes: Paths Towards Universal Reasoning Systems.

Jinlin Lai, Lixin Zou, Jiaxing Song. 2020. Optimal Mixture Weights for Off-Policy Evaluation with Multiple Behavior Policies. Offline Reinforcement Learning Workshop at Neural Information Processing Systems.

Services

- Served as a reviewer for
- ICML 2022, 2025, 2026
 - AISTATS 2023, 2024, 2025, 2026
 - AABI (ProbML) 2023, 2024, 2026
 - NeurIPS 2024, 2025, 2026
 - ICLR 2025
 - ACM SIGPLAN SLE 2025
 - TMLR

Teaching Experience

Fall 2025	Algorithms for Data Science , Teaching Assistant	<i>University of Massachusetts Amherst</i>
Spring 2025	Introduction to Computation , Teaching Assistant	<i>University of Massachusetts Amherst</i>
Fall 2024	Advanced Algorithms , Teaching Assistant	<i>University of Massachusetts Amherst</i>
Spring 2024	Probabilistic Graphical Models , Teaching Assistant	<i>University of Massachusetts Amherst</i>
Spring 2023	Probabilistic Graphical Models , Teaching Assistant	<i>University of Massachusetts Amherst</i>
Spring 2022	Probabilistic Graphical Models , Teaching Assistant	<i>University of Massachusetts Amherst</i>
Summer 2019	Algorithms for High School Olympics , Lecturer	<i>Nanchang, Jiangxi Province, China</i>
Summer 2018	Algorithms for High School Olympics , Lecturer	<i>Ganzhou, Jiangxi Province, China</i>
Summer 2017	Algorithms for High School Olympics , Lecturer	<i>Ganzhou, Jiangxi Province, China</i>

Talks

Jinlin Lai. 2023. Automatically Marginalized MCMC in Probabilistic Programming. Contributed talk in *the 5th Symposium on Advances in Approximate Bayesian Inference*.